

SVM and Conversion to Standard QP Form

1 Hard-Margin SVM Formulation

In the case of a hard-margin SVM, the goal is to find a linear classifier $b^T x + c = 0$ that perfectly separates two classes of data points, assuming that the data is linearly separable.

Given a dataset $\{(x_i, y_i)\}_{i=1}^n$, where:

- $x_i \in \mathbb{R}^n$ are the input vectors (features),
- $y_i \in \{-1, +1\}$ are the binary class labels for each data point,

the objective is to find a hyperplane, parameterized by the weight vector $b \in \mathbb{R}^n$ and the bias $c \in \mathbb{R}$, such that:

$$b^T x + c = 0$$

separates the two classes, while maximizing the margin.

The optimization problem for hard-margin SVM is defined as:

$$\begin{aligned} \min_{b, c} \quad & \frac{1}{2} \|b\|^2 \\ \text{subject to} \quad & y_i(b^T x_i + c) \geq 1 \quad \forall i = 1, 2, \dots, n \end{aligned}$$

where:

- The objective function $\frac{1}{2} \|b\|^2$ minimizes the norm of the weight vector, which maximizes the margin between the two classes.
- The constraints $y_i(b^T x_i + c) \geq 1$ ensure that all data points are correctly classified, with a margin of at least 1.

2 Conversion to Standard QP Form

We aim to convert the above problem into the standard form used by QP solvers such as CVXPY:

$$\begin{aligned} \min_x \quad & \frac{1}{2} x^T P x + q^T x \\ \text{subject to} \quad & Gx \leq h \end{aligned}$$

where $x = \begin{bmatrix} b \\ c \end{bmatrix} \in \mathbb{R}^{n+1}$, combining both the weight vector b and bias c into a single vector.

1. Objective Function

The original objective function $\frac{1}{2}\|b\|^2$ can be written in terms of x :

$$\frac{1}{2}x^T Px$$

where $P \in \mathbb{R}^{(n+1) \times (n+1)}$ is:

$$P = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix}$$

with I being the identity matrix of size $n \times n$, corresponding to the weights, and the zero entries corresponding to the bias term c .

The linear term $q^T x$ is zero because there is no linear term in the original objective, so:

$$q = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{R}^{n+1}.$$

2. Constraints

The constraint $y_i(b^T x_i + c) \geq 1$ can be written as:

$$y_i \left(\begin{bmatrix} x_i \\ 1 \end{bmatrix}^T \begin{bmatrix} b \\ c \end{bmatrix} \right) \geq 1$$

or equivalently:

$$y_i \begin{bmatrix} x_i \\ 1 \end{bmatrix}^T x \geq 1 \quad \forall i = 1, 2, \dots, n$$

This is a linear inequality of the form:

$$Gx \leq h$$

where:

- $G \in \mathbb{R}^{m \times (n+1)}$ is constructed as:

$$G = \begin{bmatrix} -y_1 \begin{bmatrix} x_1 \\ 1 \end{bmatrix}^T \\ -y_2 \begin{bmatrix} x_2 \\ 1 \end{bmatrix}^T \\ \vdots \\ -y_m \begin{bmatrix} x_m \\ 1 \end{bmatrix}^T \end{bmatrix}$$

- $h \in \mathbb{R}^m$ is a vector of -1 's:

$$h = \begin{bmatrix} -1 \\ -1 \\ \vdots \\ -1 \end{bmatrix}.$$

3 Final Problem in Standard QP Form

The final hard-margin SVM problem in standard QP form is:

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^T Px + q^T x \\ \text{subject to} \quad & Gx \leq h \end{aligned}$$

where:

- $P = \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{(n+1) \times (n+1)},$
- $q = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{R}^{n+1},$
- $G = \begin{bmatrix} -y_1 \begin{bmatrix} x_1 \\ 1 \end{bmatrix}^T \\ -y_2 \begin{bmatrix} x_2 \\ 1 \end{bmatrix}^T \\ \vdots \\ -y_m \begin{bmatrix} x_m \\ 1 \end{bmatrix}^T \end{bmatrix} \in \mathbb{R}^{m \times (n+1)},$
- $h = \begin{bmatrix} -1 \\ -1 \\ \vdots \\ -1 \end{bmatrix} \in \mathbb{R}^m.$

4 Soft-Margin SVM Formulation

In a soft-margin SVM, slack variables $\xi_i \geq 0$ are introduced to allow misclassification and penalize those points that violate the margin. The goal is to find a hyperplane that separates the two classes with a margin while allowing for some margin violations to handle non-separable data.

Given a dataset $\{(x_i, y_i)\}_{i=1}^n$, where:

- $x_i \in \mathbb{R}^n$ are the input vectors (features),
- $y_i \in \{-1, +1\}$ are the binary class labels for each data point,

the objective is to find a hyperplane, parameterized by the weight vector $b \in \mathbb{R}^n$ and the bias $c \in \mathbb{R}$, such that:

$$b^T x + c = 0$$

separates the two classes while minimizing classification errors.

Optimization Problem:

The soft-margin SVM optimization problem is defined as:

$$\begin{aligned} \min_{b,c,\xi} \quad & \frac{1}{2} \|b\|^2 + C \sum_{i=1}^n \xi_i \\ \text{subject to} \quad & y_i(b^T x_i + c) \geq 1 - \xi_i, \quad \forall i = 1, 2, \dots, n \\ & \xi_i \geq 0, \quad \forall i = 1, 2, \dots, n \end{aligned}$$

where:

- The term $\frac{1}{2} \|b\|^2$ maximizes the margin between the two classes.
- The term $C \sum_{i=1}^n \xi_i$ penalizes misclassification, where C is a regularization parameter controlling the tradeoff between maximizing the margin and minimizing classification error.
- The constraint $y_i(b^T x_i + c) \geq 1 - \xi_i$ ensures that the points are correctly classified with slack ξ_i allowing some margin violations.

5 Conversion to Standard QP Form

We aim to convert the above problem into the standard form used by QP solvers such as CVXPY:

$$\begin{aligned} \min_x \quad & \frac{1}{2} x^T P x + q^T x \\ \text{subject to} \quad & G x \leq h \end{aligned}$$

where $x = \begin{bmatrix} b \\ c \\ \xi \end{bmatrix} \in \mathbb{R}^{n+1+m}$, combining both the weight vector b , bias c , and slack variables ξ_i into a single vector.

1. Objective Function

The original objective function $\frac{1}{2} \|b\|^2 + C \sum_{i=1}^n \xi_i$ can be written in terms of x :

$$\frac{1}{2} x^T P x + q^T x$$

where: - **P matrix**: The term $\frac{1}{2} \|b\|^2$ can be represented as:

$$P = \begin{bmatrix} I & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{(n+m+1) \times (n+m+1)}$$

with I being the identity matrix of size $n \times n$, corresponding to the weights, and zeros for the bias and slack terms. - **q vector**: The linear term $C \sum_{i=1}^n \xi_i$ contributes to the q vector, where q is:

$$q = \begin{bmatrix} 0 \\ 0 \\ C \cdot \mathbf{1}_n \end{bmatrix} \in \mathbb{R}^{n+m+1}$$

where $\mathbf{1}_n$ is a vector of ones of size n .

2. Constraints

The constraint $y_i(b^T x_i + c) \geq 1 - \xi_i$ can be written as:

$$y_i \left(\begin{bmatrix} x_i \\ 1 \\ 0 \end{bmatrix}^T \begin{bmatrix} b \\ c \\ \xi \end{bmatrix} \right) \geq 1$$

or equivalently:

$$y_i \begin{bmatrix} x_i \\ 1 \\ -1 \end{bmatrix}^T x \geq 1$$

This is a linear inequality of the form:

$$Gx \leq h$$

where: - **G matrix**: We define $G \in \mathbb{R}^{(m+n) \times (n+1+m)}$ to include both the misclassification and non-negativity constraints:

$$G = \begin{bmatrix} -y_1 \begin{bmatrix} x_1 \\ 1 \\ -1 \end{bmatrix}^T \\ -y_2 \begin{bmatrix} x_2 \\ 1 \\ -1 \end{bmatrix}^T \\ \vdots \\ -y_m \begin{bmatrix} x_m \\ 1 \\ -1 \end{bmatrix}^T \\ 0 \quad 0 \quad I_\xi \end{bmatrix}$$

- **h vector**: The vector $h \in \mathbb{R}^{(m+n)}$ is a column vector of -1 's for the classification constraints and 0 's for the non-negativity constraints on ξ_i :

$$h = \begin{bmatrix} -1 \\ -1 \\ \vdots \\ -1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

6 Final Problem in Standard QP Form

The final soft-margin SVM problem in standard QP form is:

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^T Px + q^T x \\ \text{subject to} \quad & Gx \leq h \end{aligned}$$

where:

- $P = \begin{bmatrix} I & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{(n+1+m) \times (n+1+m)},$
- $q = \begin{bmatrix} 0 \\ 0 \\ C \cdot \mathbf{1}_n \end{bmatrix} \in \mathbb{R}^{n+1+m},$
- $G = \begin{bmatrix} -y_1 \begin{bmatrix} x_1 \\ 1 \\ -1 \end{bmatrix}^T \\ -y_2 \begin{bmatrix} x_2 \\ 1 \\ -1 \end{bmatrix}^T \\ \vdots \\ -y_m \begin{bmatrix} x_m \\ 1 \\ -1 \end{bmatrix}^T \\ 0 \quad \quad \quad 0 \quad I_\xi \end{bmatrix} \in \mathbb{R}^{(m+n) \times (n+1+m)},$
- $h = \begin{bmatrix} -1 \\ -1 \\ \vdots \\ -1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{(m+n)}.$